LINFO2145 — Cloud Computing

Introduction and practical information

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What is "Cloud Computing" for you?

- Let's brainstorm: add (or vote for) keywords that you would naturally associate with "cloud computing"
  - These can be technologies, methods, products, concepts, etc.
  - Max 18 chars per keyword
  - You can vote on previous proposals, but be creative and propose new ones too!

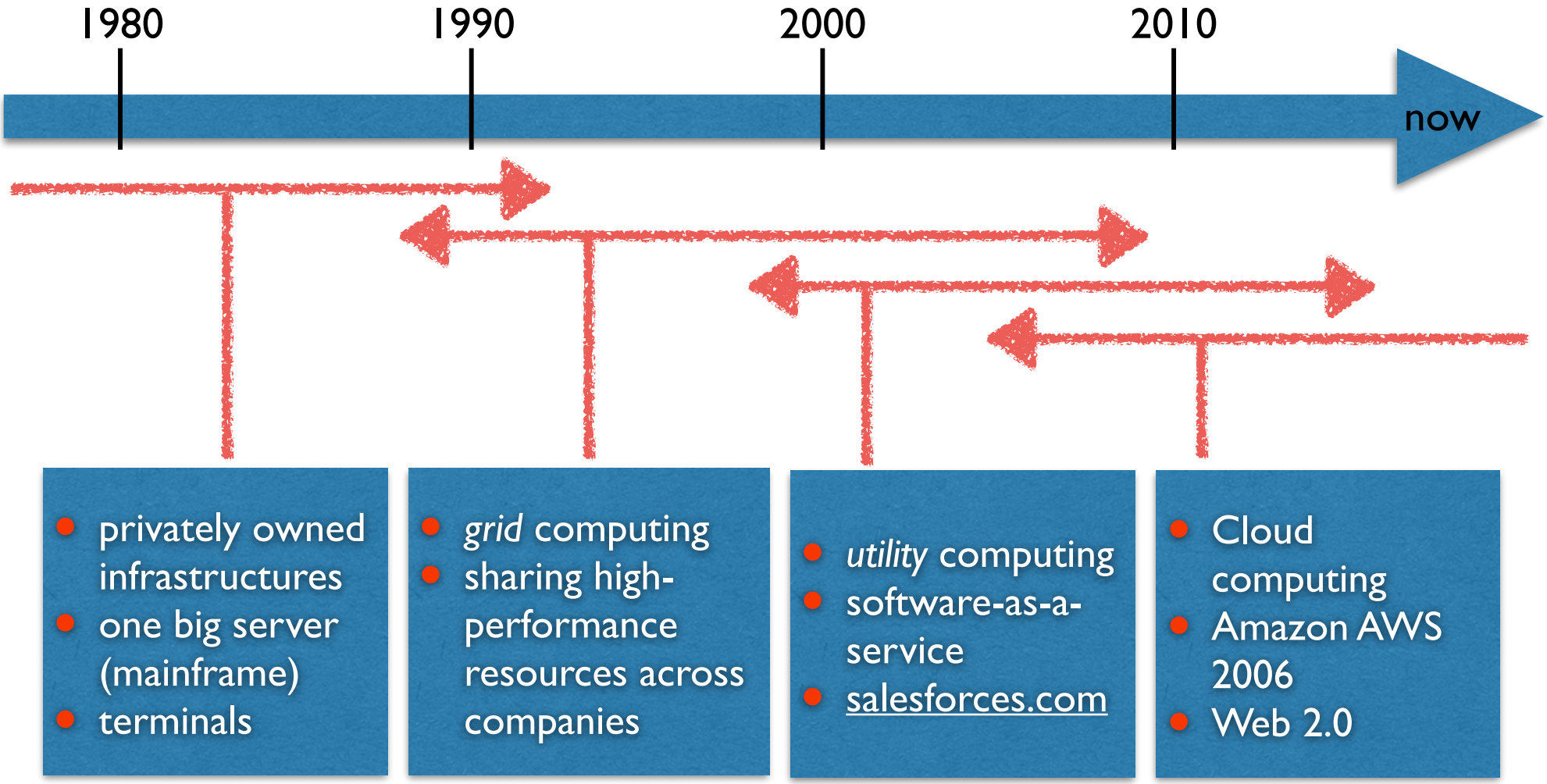
https://app.wooclap.com/MKTCEJ

or: Send @MKTCEJ to 0460 200 711

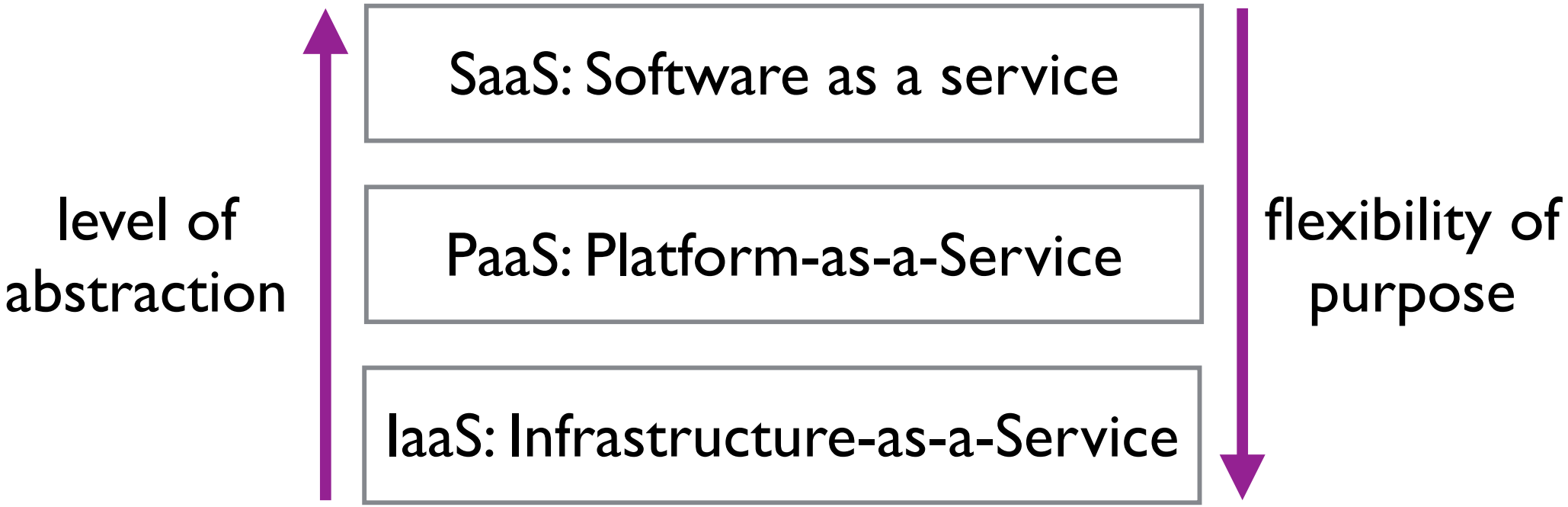
Cloud computing: a definition?

- Computing services provided as externalized resources over the Internet
  - A delivery model
  - With a large range of scenarios

The dawn of the Cloud



SPI: the 3 main cloud service models



Software-as-a-Service (SaaS)

- "Ready-to-wear" application hosted by a vendor on the cloud
  - Target: clients accessing app from a mobile client or web browser
- The main cloud computing market
- Advantages
  - Little or no administration (automatic updates)
  - Access from anywhere
  - Compatibility
  - Collaborative applications
- Disadvantages: vendor lock-in, high costs
- Target: end users

Platform-as-a-Service (PaaS)

- Hosting of managed operating system, associated environment, and support services by a vendor
  - Databases
  - Web servers
  - Identity management
  - Programming environment and libraries
- Target: application developers
- Advantage: focus on development
- Disadvantage: platform lock-in, costs when the application becomes popular

Infrastructure-as-a-Service (IaaS)

- Hosting servers and resources
- Target: IT infrastructure managers, developers
- Provides virtualized equivalent of IT resources
  - Virtual machines (servers)
    - Deployment of virtual machines images
    - Shared or single-VM physical machine
    - Alternative: deployment of containers (OS-level virtualization)
  - Storage (objects, FS, DB, raw, …)
  - Network (VLANs, …)
- Advantages: flexibility, full control, privacy
- Disadvantages: maintenance, complexity

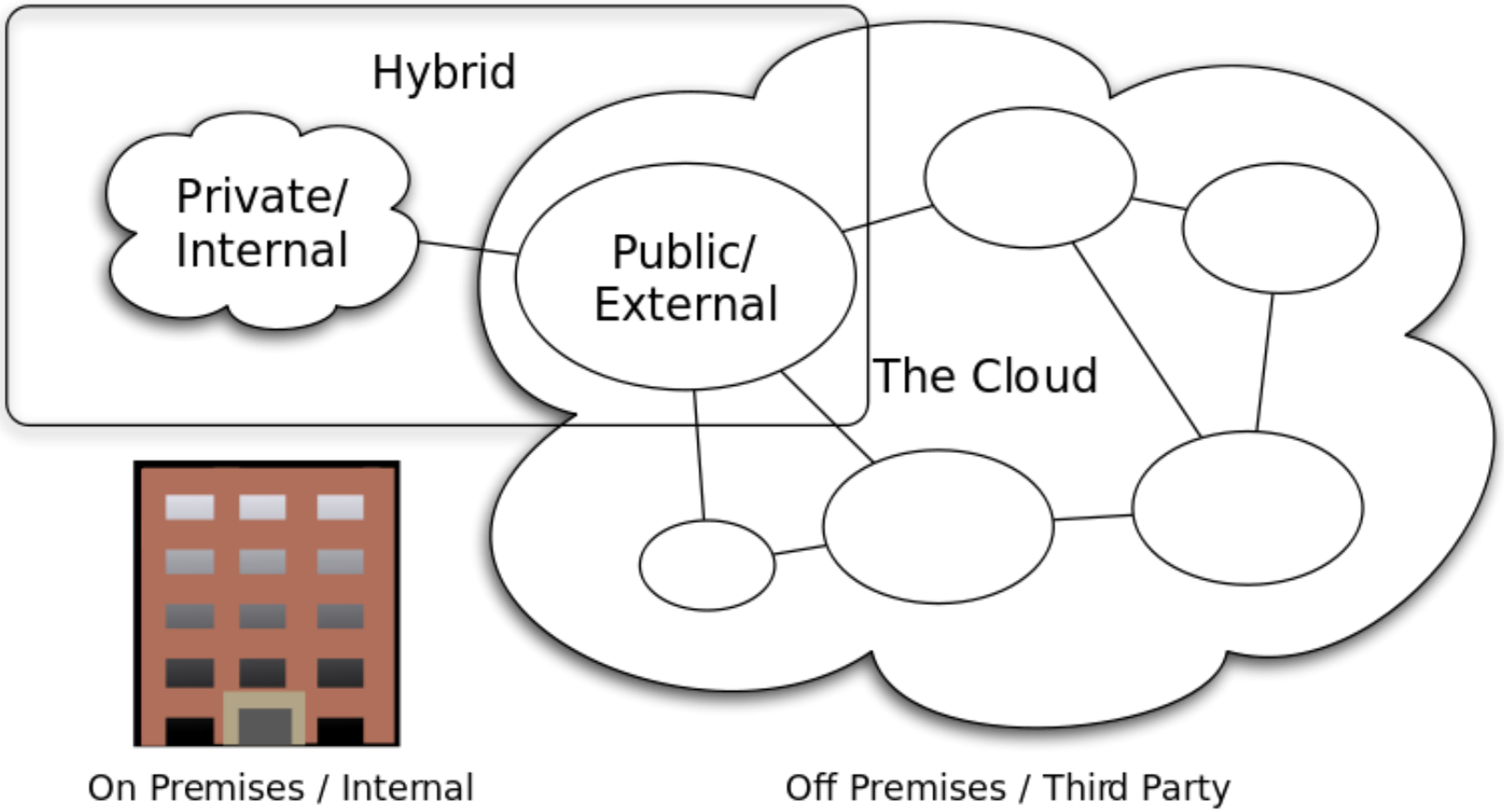
Market shares

Worldwide Public Cloud Services Revenue and Year-over-Year Growth, Calendar Year 2023 (revenues in US$ billions)

Deployment Category	2023 Revenue	2023 Market Share	2022 Revenue	2022 Market Share	2023/2022 Growth
IaaS	$133.4	19.9%	$115.5	20.7%	15.6%
PaaS	$123.3	18.4%	$95.4	17.1%	29.3%
SaaS – Applications	$298.5	44.6%	$254.4	45.6%	17.4%
SaaS – System Infrastructure Software	$114.0	17.0%	$93.1	16.7%	22.5%
Total	**$669.2**	**100%**	**$558.3**	**100%**	**19.9%**

Source: IDC Worldwide Semiannual Public Cloud Services Tracker, 2H 2023, May 2024.

Cloud deployment models



Cloud Computing Types

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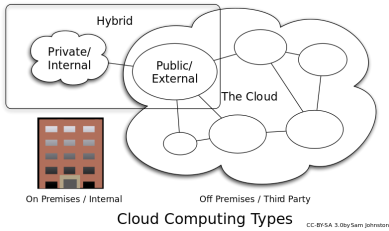
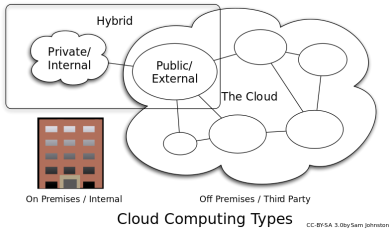
Public Clouds



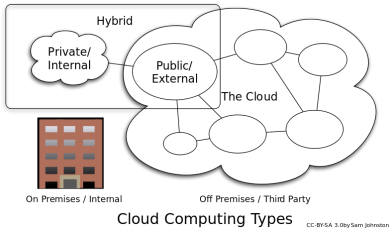
- Services rendered on a data center owned by a third-party company (Amazon, Rackspace,…)
- Over a public network
- Infrastructure shared with other clients
  - Pay-per-use model
  - No infrastructure maintenance and management costs
- Security and privacy considerations
  - Leakage of sensitive data due to attacks?
  - Legal considerations: is the NSA allowed to access my files?
  - Need to trust cloud provider with security enforcement

Private Clouds



- Services offered by an infrastructure hosted by and for a single company
  - Can avoid using public networks
  - Infrastructure shared only within the company
  - Can be installed, managed, operated by provider (e.g., Nutanix)
- Infrastructure costs for the company even when unused
- Can process private and sensitive company data with less security concerns
  - Control on where the data is stored and processed
- Same technical stack and principles as public clouds!

Hybrid Clouds

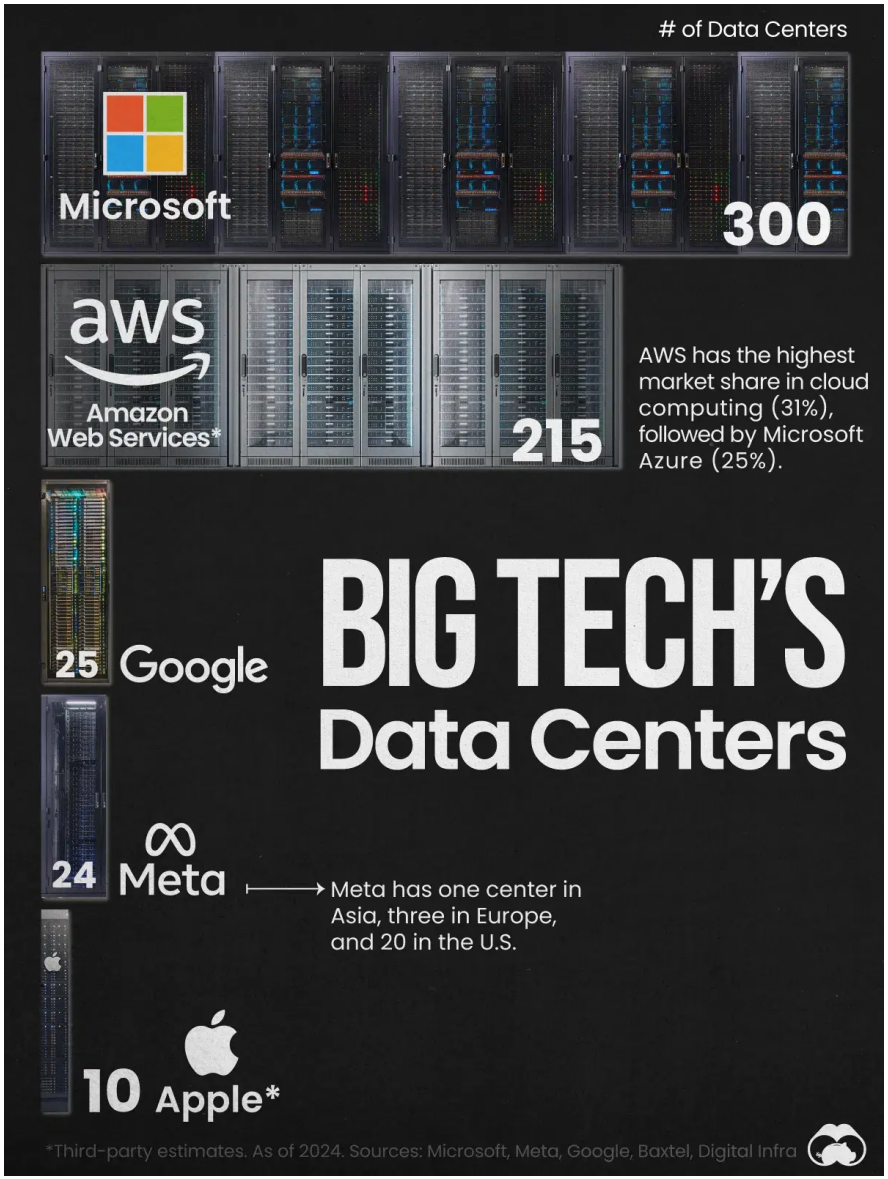


- Composition of private and public clouds
  - Multi-site deployment model
  - Combine resources from multiple providers
    - Services available only in one provider
    - Cost effectiveness
- Cloud *bursting*: temporary capacity offloading from private cloud to public cloud (reduces need to over-provision)
- Hybrid processing of sensitive/non-sensitive data
- But operation and integration more complex

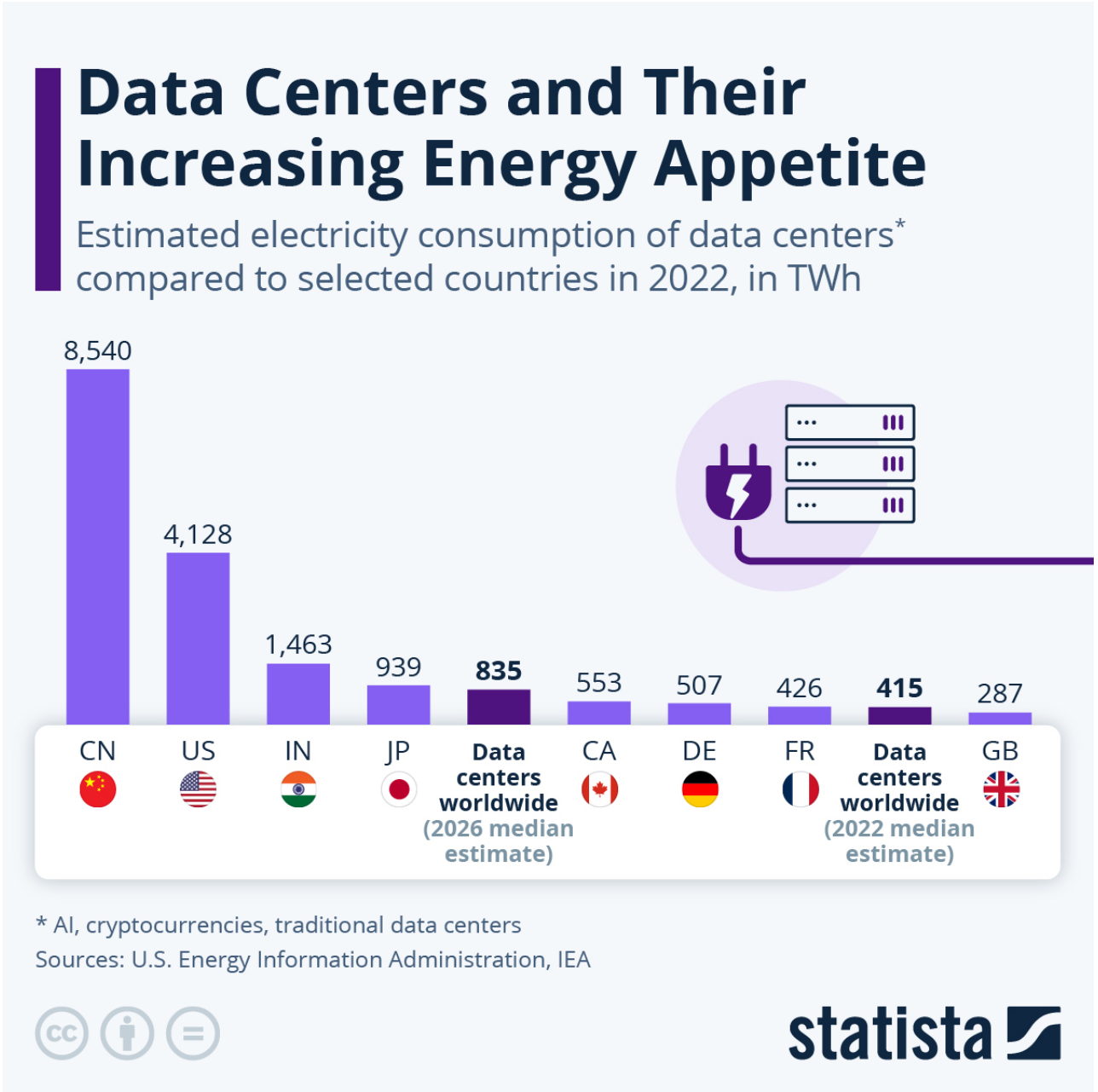
Cloud infrastructure

images source: Google inc.

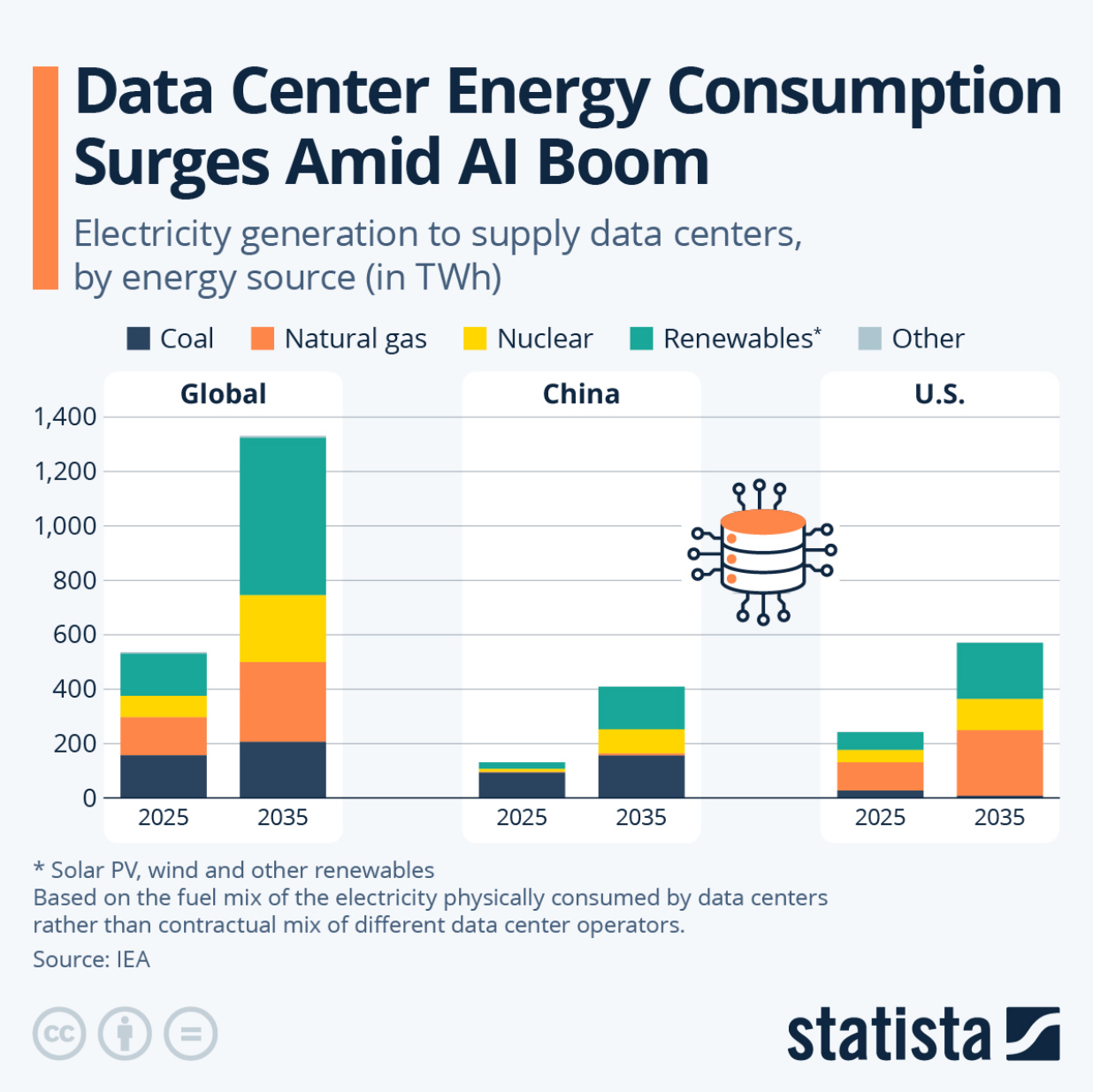
Cloud infrastructure: Data centers



If the Cloud was a country



Projections (AI-led, mostly)



google.com/datacenters

google.com/datacenters

e.com/datacenters

VD-4
Unlocked
CAP A
Unlocked
CAP B
Library Active
Service Required
google.com/datacenters

google.com/datacenters

google.com/datacenters

google.com/datacenters

Overview of the course

Objectives of the course

- Learn how to program large applications using dematerialized computing infrastructures
  - From *principles* to *applications*
- Understand software engineering as well as systems aspects
  - Cloud resource and infrastructure management
  - Service-oriented architectures, data stores, security, etc.
- Focus #1: Handling lots of users
  - How to build a reliable & fast application in the Cloud?
- Focus #2: Handling lots of data
  - How to make use of the massive data that users generate?

Part A: enabling technologies and infrastructure management

- Virtualization: key technology for Cloud computing
  - Share a physical resource (machine, network) between multiple tenants
  - Separate the management of the machine from the management of the operating system
  - Separate the management of the operating system from the management of the applications

➡ Two main forms

  - Machine virtualization (today)
  - OS-level virtualization and containers (also today)

\+ Advanced topics on IaaS systems and management

Part B: Building Scalable Applications

- How can one build an application in the Cloud to support many users?
  - A 'traditional' single-machine application won't work
- Collections of small, independent services that can be replicated or shut down, depending on the load
  - Micro-services and serverless computing
  - Elastic scaling and energy efficiency
- Databases adapted to Cloud scale
  - Replication and availability
  - Tradeoffs between consistency and performance

Part C - Big Data Processing

- Large amounts of user-generated data
  - Purchase history, comments, navigation history, etc.
  - Mobile: GPS location, periods of activity, contacts, etc.
  - Application monitoring
- Derive value: personalized ads, marketing, etc.
  - If it's free, you're the product —
    "*Si c'est gratuit, c'est toi le produit*"
- Make service better, e.g., Netflix recommendations
- Need processing paradigms adapted to Cloud scale
  - Volume: Map/Reduce paradigm
  - Velocity: Stream processing

Advanced topics

- Tentative list for this year
  - Cloud coordination: how to build fault-tolerant Cloud services and applications
  - Cloud and DevOps practices
  - Security and privacy
    - Addressing confidentiality concerns in the Cloud
  - Blockchains, distributed trust, decentralized clouds
  - Possibly a guest industry presentation

Practical information

Agenda

- Lessons
  - Wednesdays 10:45-12:45
  - In BARB93, SUD11 on 22/10 (check ade-scheduler)
- Practical sessions
  - Fridays 14:00 to 16:00
  - Always in the Intel room (+ adjacent if necessary)
- Announcements and information on Moodle

Semester project

- Building a modern application: a web store
  - Front-end provided, but using only local storage
  - Your goal will be to build the Cloud *backend* for this application (part #1) …
  - … and to leverage Big Data processing to take advantage of the data generated by its users (part #2)
- Objective: gradually use and practice concepts seen in class in your project

➡ Teams of two students

  - Selection on Moodle before the start of the project

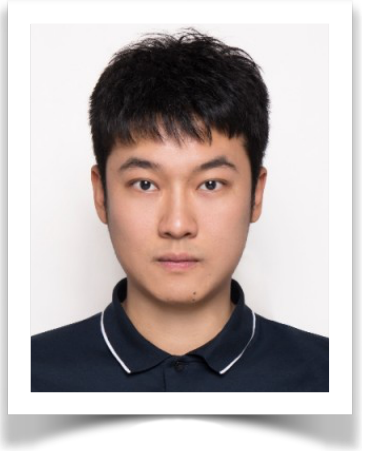
Tutorials

- *Bootstrap* tutorial
  - Using virtual machines and docker containers
- Several tutorials during the project
  - Project set up, use of cloud database, etc.
  - Optional tutorial on Kubernetes
- Tutorials are not graded

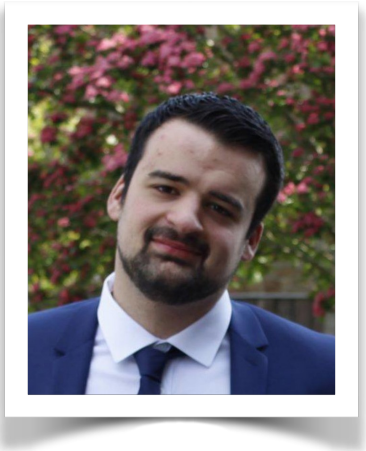
- Solo (recommended for bootstrap) or in groups of two

Team and contacts

- Office hours *by email appointment*
  - etienne.riviere@uclouvain.be
- General questions on the course or on the assignments
  - Use the Moodle forum if the answer can be useful for others
  - Always contact *all three* assistants when sending an email
  - yinan.cao@uclouvain.be ; romain.carlier@uclouvain.be ; donatien.schmitz@uclouvain.be
- Please do not send direct Teams messages



Yinan



Clément

Paul-Édouard

Online quizzes

- Every 1 or 2 weeks: quiz available on Moodle
  - 5 to 6 questions
  - About the lecture and practical sessions
  - Answered individually
- Peer-review process: each quiz is graded/commented on by *two* other students
- Answering quizzes and providing feedback accounts for 10% of the final grade
  - 60% for your answers, 40% for grading
  - Note: this is a participation grade, the grade provided by other students is only indicative
  - If you do not receive feedback (bad luck) ask the TAs to provide you with one (this will not impact your grade)

Practical considerations

- We assume everyone uses her/his own laptop
  - If you do not have one, let us know

- Cloud resources: Your UCLouvain account gives you access to credits on Microsoft Azure
  - We will start with a mini private-cloud running on your own laptop
  - Some credit also available with Github student developer pack (details on Moodle); we recommend asking for it too

GitHub

- Tutorials and project bootstrap code will be available on GitHub
  - Requires the ability to create/fork private repositories
  - Suggested (but not mandatory): Request GitHub student developer pack
- Provide us with your GitHub identifier in the Moodle database
  - Do not forget to provide your *real* name as well
- For ULB/external students
  - If no access to Moodle, give your identifier directly to the assistants
  - We will post the slides on GitHub if necessary, but we cannot post the quizzes (sorry)

Grading and Evaluation

- First session
  - Project 45%
  - Final exam 45%
  - Quizzes (participation) 10%

- Second session
  - It is not possible to redo the project or the quizzes for the second session
  - Quizzes do not count in the second session
  - Project (from first session) 45%
  - Final exam 55%

Rules regarding use of AI

- Complete rules at https://www.uclouvain.be/en-cours-2025-linfo2145
- Use is **prohibited** for online quizzes (answering and peer review)
  - Participation is graded, not quality. Letting AI respond is just useless
  - We will perform random checks, and if we find blatant use of AI, we can request an oral examination and verify genuineness (risk: 0 for all quizzes)
- AI for the project (code help) is **discouraged** but **not prohibited**, following the observation of mandatory rules:
  - You must take full responsibility for the work and be able to explain any line of code or documentation in an oral examination
    - We are free to request such an oral examination at any time
  - Any use of AI for the project (code, documentation) must be properly documented in the specific section of the project documentation
  - Failure to document AI use will be considered plagiarism
  - According to EPL rules, abusive AI use undermining the acquisition of knowledge targeted by the project may be considered an irregularity and reported to the president of the juries, for investigation and decision

General comments

- Feedback and constructive comments highly appreciated
  - Let us know early if you face issues with the tutorials, practicals, or quizzes
- Be active and participate
  - Ask questions during the lectures
  - Use the Moodle forum
- Tutorials and projects on GitHub
  - Send us a *pull request* if you spot an error or would like to add a clarification
  - Can grant bonus points in the project for significant contributions

Schedule (tentative)

Cloud Computing -- Fall 2025

			LECTURES		PRACTICALS	
Week		Lecture	Wednesdays 10:45-12:45	Session	Fridays, 14:00-16:00	Project
1	17-Sep-25	1	**Introduction (30 minutes), Virtualization (1h30)**			
1	19-Sep-25			1	**Tutorial 1: setup/ containers**	
2	24-Sep-25	2	**PaaS and SOA**			
2	26-Sep-25			2	**Tutorial: Project setup**	**Projet part A**
3	1-Oct-25		*no lecture*			
3	3-Oct-25			3	**setup check**	
4	8-Oct-25	3	**Storage and Data Management**			
4	10-Oct-25			4	**Tutorial: DB**	
5	15-Oct-25	4	**Elasticity and Energy**			
5	17-Oct-25			5	**Tutorial: Elasticity**	
6	22-Oct-25	5	**Fault tolerance, Geo-replication and Consistency**			
6	24-Oct-25			6	**mentoring**	
7	29-Oct-25		*no lecture, SMART week*			
7	31-Oct-25				*no practical, SMART week*	
8	5-Nov-25	6	**Fault tolerance, ZooKeeper**			
8	7-Nov-25			7	**Tut.: Azure**	
9	12-Nov-25	7	**Advanced topics: systems aspects of IaaS**			
9	14-Nov-25			8	**mentoring**	
10	19-Nov-25	8	**Advanced topics: security**			
10	21-Nov-25			9	**mentoring**	
11	26-Nov-25	9	**Big Data processing with Map/Reduce**			
11	28-Nov-25			10	**Tutorial: M/R w/ CouchDB**	**Project part B**
12	3-Dec-25	10	**Big Data beyond Map/Reduce**			
12	5-Dec-25			11	**mentoring**	
13	10-Dec-25	11	**Advanced topics: Blockchains**			
13	12-Dec-25			12	**mentoring**	
14	17-Dec-25	12	*Guest speaker (TBC)*			
14	19-Dec-25				**mentoring**	
	Exam period		**Exam date in January**			

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